TED (15) 3014 Reg. No: ………………..

(REVISION-2015) Signature………………

MODEL QUESTION PAPER

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING TECHNOLOGY

BUILDING PLANNING AND DRWING

(MAX.MARK:100)

[Note:-1,Question No.II is compulsory.

2,Missing data can be suitably assumed.

3,A2 size drawing sheet to be supplied]

PART-A

(Max.Mark:15)

(Time:3hours)

I Answer the following questions in one or two sentences .Each question carries 1 ½ marks.

1. Define the term Building.
2. List the classification of residential building as per NBC.
3. Specify the requirements of plot for an Educational building as per KMBR.
4. Define FAR.
5. Specify the minimum depth of foundation and plinth height of a building as per KMBR.
6. Write the size for tread and rise of a stair in a public building.
7. Differentiate between Hip rafter and Valley rafter.
8. Draw the conventional signs of brick, steel and wood.
9. List the documents to be submitted for obtaining building permit.
10. Specify the minimum area of opening for lighting and ventilations in different Climates. (10x 1 ½ = 15)

PART-B

(Max.Mark:85)

II (a) Prepare the line plan for a residential building to suit for a plot of 25x 22m size based on the rules and regulations of KMBR.The total built up area of the building is 183.50m^2 and should contain the following facilities

(i) Drawing and dining

(ii) bed rooms -2nos,with attached toilets (iii)Kitchen and pantry

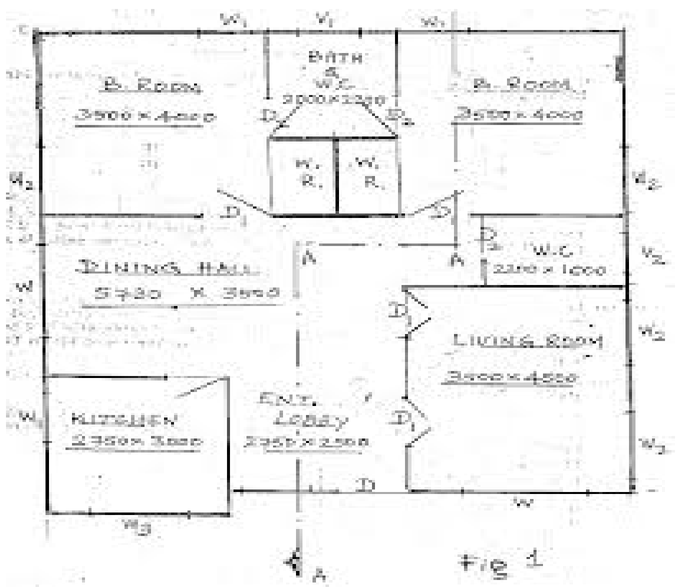
(iv) Verandah

(V) Porch. etc.

A road 6m width passing along the 22m side of the plot,which is in the east direction

25

(b) A



The line plan in Fig 1 gives the arrangement of room of a residential building. Prepare its

(i) plan (ii) section along AA

Details

(i) Bed concrete for foundation,PCC, 1:4:8- 80cmx20cm

(ii) RR masonry in cm 1:8 for foundation and basement– 60cmx60cm a and 45cm x45cm.

(iii)Superstructure, brick wall in cement mortar 1:6,22 cm thick, 360cm height for tiled portion and 300cm for terraced portion.

(iv)RCC M20 grade, 12cm thickness.

(v) Details of tiled roofing.

Roofing with MP tiles,

Ridge- 5cmx15cm.

Common rafter - 7cmx10cm.

Collar- 7cmx10cm.

Wall plates-10cmx7cm

Eave board—1.5cmx12cm

All other data such as position and size of doors, windows, ventilators etc. Can suitably be assumed. (15 + 15 =30)

(III) Draw to a suitable scale the elevation and sectional plan of a panelled door with given details.

Size of door – 90cm x 210cm, 2 leaves

Door frame – 9cm x 7cm.

Style – 9.5cm x 3.5 cm.

Top rail – 9.5cm x 3.5 cm.

Sash bar – 3.5 cm x 3.5 cm.

Lock rail – 19.5 cm x 3.5 cm.

Bottom rail – 19.5 cm x 3.5 cm.

Panel – 1.6 cm thick (15)

OR

IV Draw the elevation of a single collar roof showing all details:

Wall thickness – 30cm.

Clear span – 5m.

Collar – 4cm x 12.5 cm.

Ridge – 8cm x 20 cm.

` Rafter – 5 cm x 12.5 cm.

Eave projection – 60 cm

Wall plates – 15 cm x 10 cm. (15)

V Draw the half sectional elevation along the centre line of the road of a slab culvert designed for a highway across a stream with the following details:

Clear span – 3.0m.

` Abutment in PCC 1:3:6, trapezoidal section, water face vertical.

Top width 60cm,bottom width 120 cm, foundation PCC 1:3:6,180cm wide and 60cm deep.

Square return walls 400cm length from the earth face of abutment top.

Water way lined 20cm thick, stone pitching.

Bed slab 30cm thick RCC,M20 grade, with bearing of 40cm on abutment.

Provide parapets and railing etc.suitably.

Top level of culvert slab -- +100.00m.

Top level of stone pitching -- +97.24m.

RL of base of foundation of abutment -- +95.85m (15)

OR

VI Prepare a plumbing layout of (Fig. I) showing the water supply and sanitary fixtures. (15)
